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CYBER SECURITY PROJECT: SCANNING NETWORKS

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Topic: Automated Port Scanner and Web Dashboard Generation

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Operating System: Kali Linux

Tools Used: Nmap, Netdiscover, Masscan, Apache2, Bash, Cron

Date: July 21, 2026 Status: Completed & Verified

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1. PROJECT OVERVIEW & DESCRIPTION

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1.1 Description

Port scanning and network discovery are essential components of security testing and penetration testing. Building and using a custom automated scanner leveraging multiple network reconnaissance utilities (Netdiscover, Masscan, and Nmap) with structured output generation is a fundamental skill for any security professional.

Without proper scanning tools, security analysts cannot discover critical information about the network and the nodes attached to it. Multi-layered discovery tools help track all connected devices, making it possible to identify unauthorized or suspicious devices running on the network.

1.2 The Core Problem & Solution

Network administrators and pentesters cannot monitor a network manually at all times. A manual scan only captures connected devices at that specific moment; devices that connect and disconnect between manual scans go completely unnoticed.

To solve this challenge, this project implements a multi-tool automated strategy:

1. Netdiscover runs rapid layer-2 ARP discovery to identify active IP/MAC addresses.
2. Masscan performs high-speed port sweeps on identified targets.
3. Nmap collects detailed service version information.
4. Apache2 serves all combined scan data live on a local web dashboard refreshed every 10 minutes via Linux cron jobs.

1.3 Tools & Environment

- * Netdiscover: Layer-2 ARP discovery utility for fast local subnet host mapping.
- * Masscan: Ultra-high-speed port scanner used to rapidly identify open ports across active hosts.
- * Nmap: Detailed service port auditor used for deep inspection of active ports.
- * Apache2: Local HTTP web server hosting the live scan results dashboard.
- * Cron Daemon: Linux system service executing background scheduled tasks every 10 minutes.

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2. STEP-BY-STEP APPROACH & IMPLEMENTATION

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2.1 Step 1: Environment & Tool Installation

The required scanning and web hosting utilities were verified and installed on the Kali Linux environment using the system package manager:

Commands Executed:

```
sudo apt update  
sudo apt install nmap netdiscover masscan apache2 -y
```

Tool References:

- * Nmap: <https://nmap.org/download>
- * Masscan: <https://github.com/robertdavidgraham/masscan>
- * Netdiscover: <https://github.com/netdiscover-scanner/netdiscover>
- * Apache2: <https://httpd.apache.org/download.cgi>

2.2 Step 2: Apache2 Web Server Configuration

The Apache2 HTTP server was configured and enabled to serve the dynamic scan results on the local network at <http://localhost/index.html>.

Commands Executed:

```
sudo systemctl start apache2  
sudo systemctl enable apache2
```

```
sudo systemctl status apache2
```

2.3 Step 3: Script Creation & Multi-Tool Integration

A custom shell script (`network_scanner.sh`) was constructed to run Netdiscover for host discovery, Masscan for fast port sweeps, and Nmap for service auditing against the target subnet (`10.0.2.0/24`), automatically attach a live execution timestamp, and format the output directly into `/var/www/html/index.html`.

Script Source Code (`/home/kali/network_scanner.sh`):

```
#!/bin/bash

OUTPUT_FILE="/var/www/html/index.html"

TIMESTAMP=$(date "+%Y-%m-%d %H:%M:%S")

# 1. ARP Host Discovery with Netdiscover
NETDISCOVER_OUT=$(sudo netdiscover -i eth0 -r 10.0.2.0/24 -P 2>&1)

# 2. Fast Port Sweep with Masscan
MASSCAN_OUT=$(sudo masscan 10.0.2.0/24 -p80,443,22,135,3306 --router-ip 10.0.2.2 --rate=1000
2>&1 | grep Discovered)

# 3. Service Auditing with Nmap
NMAP_OUT=$(nmap -F 10.0.2.0/24)

cat <<EOF > $OUTPUT_FILE

<!DOCTYPE html>

<html>

<head>

  <meta charset="UTF-8">

  <title>Automated Network Port Scan Results</title>

</head>

<body>

  <h1>[!] Automated Network Port Scan Results</h1>

  <p><b>System Time:</b> $TIMESTAMP</p>
```

Target Scan Scope: 10.0.2.0/24

1. Active Host Discovery (Netdiscover)

```
$NETDISCOVER_OUT
```

2. High-Speed Port Sweep (Masscan)

```
$MASSCAN_OUT
```

3. Detailed Service Scan (Nmap)

```
$NMAP_OUT
```

Kali Linux Lab Utility • Refreshes automatically every 10 minutes via Cron.

EOF

```
GNU nano 9.1 /home/kali/network_scanner.sh
#!/bin/bash
OUTPUT_FILE="/var/www/html/index.html"
TIMESTAMP=$(date +%Y-%m-%d %H:%M:%S)

# 1. ARP Host Discovery with Netdiscover
NETDISCOVER_OUT=$(sudo netdiscover -i eth0 -r 10.0.2.0/24 -P 2>&1)

# 2. Fast Port Sweep with Mass
MASSCAN_OUT=$(sudo masscan 10.0.2.0/24 -p80,443,22,135,3306 --router-ip 10.0.2.2 --rate=1000 2>&1 | grep Discover)

# 3. Service Auditing with Nmap
NMAP_OUT=$(nmap -F 10.0.2.0/24 2>&1)

cat <<EOF > $OUTPUT_FILE
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Automated Network Port Scan Results</title>
</head>
<body>
<h1>[[ Automated Network Port Scan Results</h1>
<p><b>System Time:</b> $TIMESTAMP</p>
<h3>Target Scan Scope: 10.0.2.0/24</h3>

<h2>1. Active Host Discovery (Netdiscover)</h2>
<pre>
$NETDISCOVER_OUT
</pre>

<h2>2. High-Speed Port Sweep (Masscan)</h2>
<pre>
$MASSCAN_OUT
</pre>

<h2>3. Detailed Service Scan (Nmap)</h2>
<pre>
$NMAP_OUT
</pre>

<p>Kali Linux Lab Utility - Refreshes automatically every 10 minutes via Cron.</p>
</body>
</html>
EOF
```

Permissions Granted:

```
chmod +x /home/kali/network_scanner.sh
```

2.4 Step 4: Cron Job Automation (10-Minute Schedule)

To automate execution without manual user interaction, a user crontab entry was established to run the scanner script every 10 minutes continuously.

Crontab Configuration (`crontab -e`):

```
*/10 * * * * /home/kali/network_scanner.sh
```

3. TASK VERIFICATION & EVIDENCE (PROOF OF COMPLETION)

3.1 Task 1: Background Automation Verification (Cron Execution Logs)

The background execution of the script was confirmed using system journal logs to verify recurring triggers at 10-minute intervals.

Command Executed:

```
sudo journalctl -u cron -n 20
```

```
(kali@kali)-[~]
└─$ sudo journalctl -u cron -n 20
Jul 24 09:50:01 kali CRON[126213]: (kali) CMD (/home/karthik/network_scanner.sh >/dev/null 2>&1)
Jul 24 09:50:01 kali CRON[126209]: (CRON) info (No MTA installed, discarding output)
Jul 24 09:50:01 kali CRON[126209]: pam_unix(cron:session): session closed for user root
Jul 24 09:50:01 kali CRON[126208]: pam_unix(cron:session): session closed for user kali
Jul 24 09:53:01 kali cron[598]: (kali) RELOAD (crontabs/kali)
Jul 24 09:53:01 kali cron[598]: (root) RELOAD (crontabs/root)
Jul 24 09:55:01 kali CRON[128845]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jul 24 09:55:01 kali CRON[128847]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Jul 24 09:55:01 kali CRON[128845]: pam_unix(cron:session): session closed for user root
Jul 24 10:00:01 kali CRON[131374]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jul 24 10:00:01 kali CRON[131376]: (root) CMD (/home/kali/network_scanner.sh)
Jul 24 10:00:01 kali sudo[131380]: root : PWD=/root ; USER=root ; COMMAND=/usr/sbin/netdiscover -i eth0 -r 10.
Jul 24 10:00:01 kali sudo[131380]: pam_unix(sudo:session): session opened for user root(uid=0) by root(uid=0)
Jul 24 10:00:06 kali sudo[131380]: pam_unix(sudo:session): session closed for user root
Jul 24 10:00:06 kali sudo[131434]: root : PWD=/root ; USER=root ; COMMAND=/usr/bin/masscan 10.0.2.0/24 -p80,44
Jul 24 10:00:06 kali sudo[131434]: pam_unix(sudo:session): session opened for user root(uid=0) by root(uid=0)
Jul 24 10:00:29 kali sudo[131434]: pam_unix(sudo:session): session closed for user root
Jul 24 10:00:42 kali CRON[131374]: pam_unix(cron:session): session closed for user root
Jul 24 10:05:01 kali CRON[134134]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jul 24 10:05:01 kali CRON[134134]: pam_unix(cron:session): session closed for user root
lines 1-20/20 (END)
```

Figure 1: Terminal output showing systemd cron logs confirming execution of network_scanner.sh every 10 minutes.

3.2 Task 2: Live Web Dashboard Verification (Apache HTTP Server)

The rendered webpage was verified by opening Firefox and navigating to <http://localhost/index.html>. The page successfully displays multi-tool outputs from Netdiscover, Masscan, and Nmap alongside live timestamps updated by the background script.

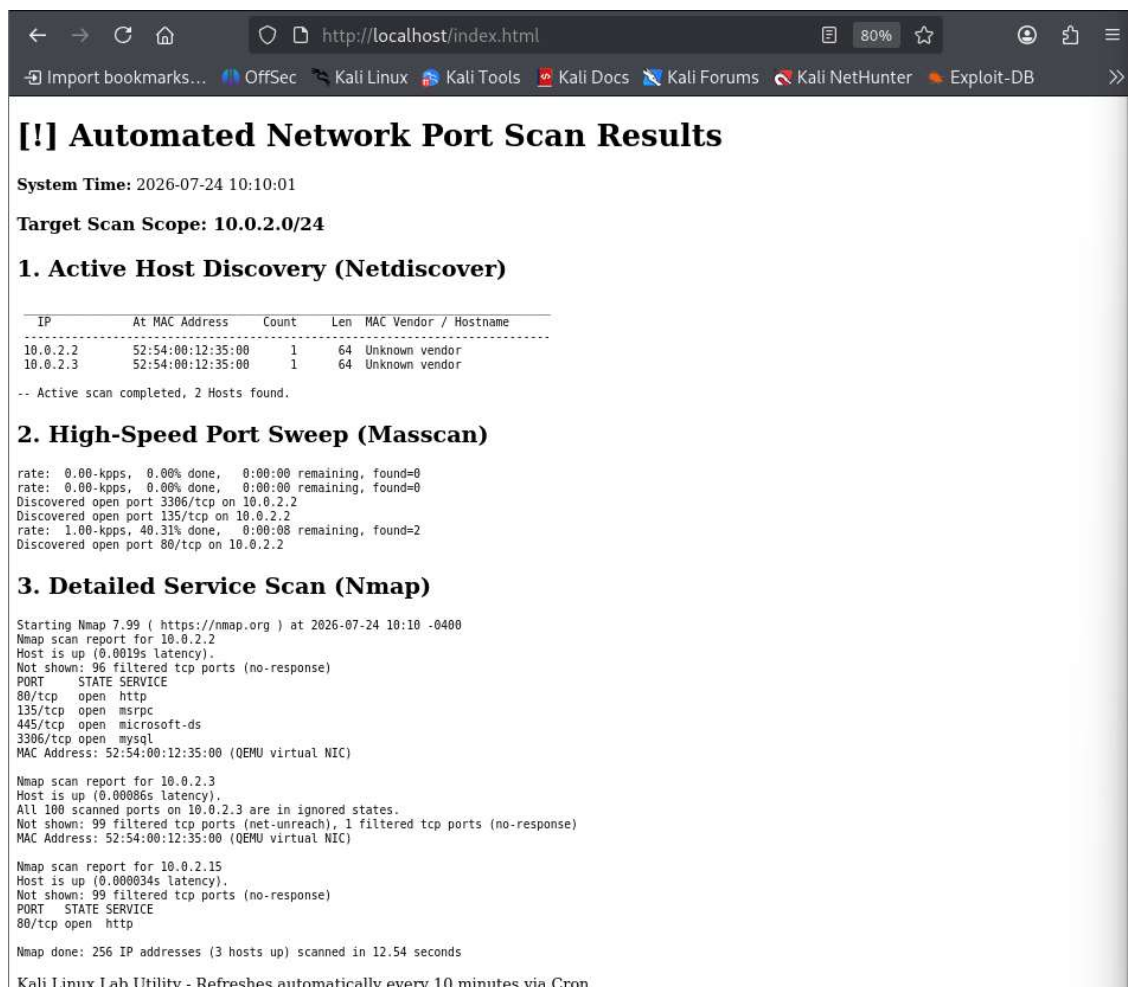


Figure 2: Browser output displaying live multi-tool network scan results hosted on Apache2 at <http://localhost/index.html>.

3.3 Task 3: Video Demonstration (Proof of Concept)

A full video demonstration showing the background cron job executing every 10 minutes and automatically updating the live Apache2 web dashboard is accessible via the link below:

* Video Link:

https://drive.google.com/file/d/1ZDvi7lBakVDQ_XGlaBPQZ2u1OYBs5GyB/view?usp=sharing

* Recorded Environment: Kali Linux Virtual Machine (10.0.2.0/24 Subnet)

4. LEARNING OUTCOMES & SUMMARY

4.1 Key Outcomes

- * Learned how to combine Netdiscover for ARP discovery, Masscan for fast port sweeps, and Nmap for detailed port scanning.
- * Successfully configured and managed an Apache2 web server service on Kali Linux.
- * Mastered Linux task automation using shell scripts and system cron daemon configuration.
- * Implemented live dynamic reporting with automatic timestamping to monitor network changes every 10 minutes.

4.2 Tasklist Checklist

- [x] Download and verify tools (Nmap, Masscan, Netdiscover, Apache2).
- [x] Configure Apache2 HTTP web server on local system.
- [x] Integrate Netdiscover, Masscan, and Nmap outputs into dynamic HTML web pages with timestamps.
- [x] Automate process execution every 10 minutes via cron job.
- [x] Verify live web dashboard access within the network.

5. CONCLUSION

This project successfully implemented an automated network scanning solution using Kali Linux. Netdiscover was used to discover active hosts on the network, Masscan performed high-speed port scanning, and Nmap provided detailed service and port information. A Bash script automated the complete scanning process, while Apache2 displayed the scan results through a web dashboard. Cron jobs ensured that the scans were executed automatically every 10 minutes without manual intervention.

This project enhanced practical knowledge of network scanning, Linux automation, web server configuration, and cybersecurity tools used in penetration testing and network administration.